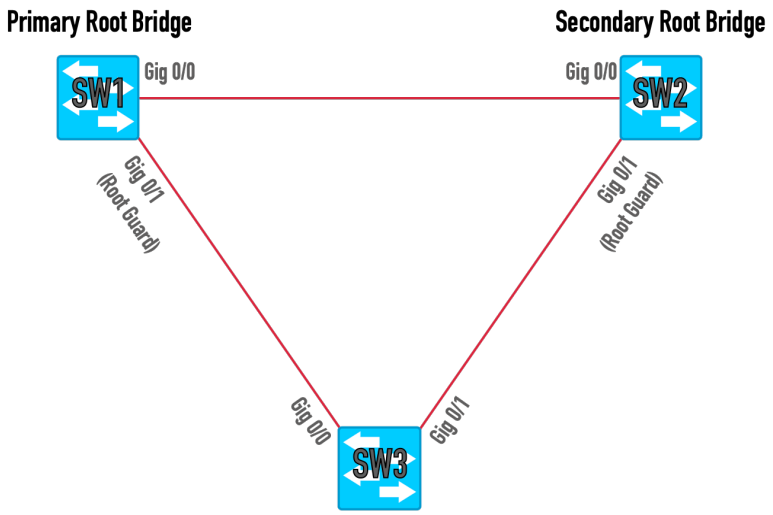
Root Guard

Topology



Initial Configuration Commands

SW1:

```
enable
conf term
hostname SW1
line con 0
exec-timeout 0 0
logging synchronous
exit
no ip domain-lookup
int gig 0/2
shutdown
exit
int gig 0/3
shutdown
exit
spanning-tree vlan 1 root primary
end
copy run star
```

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SW2:

```
enable
conf term
hostname SW2
line con 0
exec-timeout 0 0
logging synchronous
exit
no ip domain-lookup
int gig 0/2
shutdown
exit
int gig 0/3
shutdown
exit
spanning-tree vlan 1 root secondary
end
copy run star
```

SW3:

```
enable
conf term
hostname SW3
line con 0
exec-timeout 0 0
logging synchronous
exit
no ip domain-lookup
int gig 0/2
shutdown
exit
int gig 0/3
shutdown
exit
end
copy run star
```

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Lab Tasks

1. First, let's confirm that SW1 is the root by using the "show spanning-tree" command.
2. Let's go to SW2 and see what our bridge priority is there.
3. Let's go to SW3 and see what our bridge priority is there.
4. Let's activate the root guard feature on SW1 and SW2 for their Gig 0/1 interfaces. After that, let's see if we have any ports currently in the Root Inconsistent state by using the command "show spanning-tree inconsistentports" on both SW1 and SW2.
5. Let's create a scenario where a bad actor gains control of SW3 and wants it to become the root bridge. To configure this, let's change the bridge priority of Vlan 1 on SW3 to be 4096.
6. On SW1 and SW2, let's see if we have any inconsistent ports.

Solution

Step 1: First, let's confirm that SW1 is the root by using the "show spanning-tree" command.

```
SW1>en
SW1#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     5254.001c.5857
             This bridge is the root
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    24577  (priority 24576 sys-id-ext 1)
             Address     5254.001c.5857
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
----------
Gi0/0               Desg FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
```

(##We can see here that SW1 is the root bridge, and we can also see that our bridge priority is 24577.)

Step 2: Let's go to SW2 and see what our bridge priority is there.

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```
SW2>en
SW2#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     5254.001c.5857
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    28673  (priority 28672 sys-id-ext 1)
             Address     5254.0007.0714
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
----------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Desg FWD 4         128.2    P2p
```

(##We can see here that our bridge priority is 28672. This bridge priority is lower than the default bridge priority of 32768, but it is not lower than the bridge priority of our root bridge. If the root bridge was to become unavailable, then SW2 would take over as the root bridge.)

Step 3: Let's go to SW3 and see what our bridge priority is there.

```
SW3>en
SW3#show spanning-tree

VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    24577
             Address     5254.001c.5857
             Cost        4
             Port        1 (GigabitEthernet0/0)
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
             Address     5254.0016.98c0
             Hello Time   2 sec  Max Age 20 sec  Forward Delay 15 sec
```

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```
             Aging Time  300 sec

Interface           Role Sts Cost      Prio.Nbr Type
------------------- ---- --- --------- -------- ----------------------
----------
Gi0/0               Root FWD 4         128.1    P2p
Gi0/1               Altn BLK 4         128.2    P2p
```

(##We can see here that SW3 has the default bridge priority of 32768.)

Step 4: Let's activate the root guard feature on SW1 and SW2 for their Gig 0/1 interfaces. After that, let's see if we have any ports currently in the Root Inconsistent state by using the command "show spanning-tree inconsistentports" on both SW1 and SW2.

```
SW1#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW1(config)#int gig 0/1
SW1(config-if)#spanning-tree guard root
SW1(config-if)#end
SW1#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------

Number of inconsistent ports (segments) in the system : 0

SW2#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW2(config)#int gig 0/1
SW2(config-if)#spanning-tree guard root
SW2(config-if)#end
SW2#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------

Number of inconsistent ports (segments) in the system : 0
```

(##We can see on both SW1 and SW2 that we have no inconsistent ports.)

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Step 5: Let's create a scenario where a bad actor gains control of SW3 and wants it to become the root bridge. To configure this, let's change the bridge priority of Vlan 1 on SW3 to be 4096.

```
SW3#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
SW3(config)#spanning-tree vlan 1 priority 4096
SW3(config)#end
```

(##Based on the spanning-tree information that we pulled earlier from SW3, we can see that the bridge priority of our root bridge is 24577. So, to make SW3 the root bridge we had to choose a bridge priority that was going to be lower than that of our root bridge.)

Step 6: On SW1 and SW2, let's see if we have any inconsistent ports.

```
SW1#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------
VLAN0001             GigabitEthernet0/1       Root Inconsistent

Number of inconsistent ports (segments) in the system : 1

SW2#show spanning-tree inconsistentports

Name                 Interface                Inconsistency
-------------------- ------------------------ ------------------
VLAN0001             GigabitEthernet0/1       Root Inconsistent

Number of inconsistent ports (segments) in the system : 1
```

(##We can see that on SW1 we have one inconsistent port, and it is Gig 0/1. Gig 0/1 is in a Root Inconsistent state because we are receiving superior BPDUs off of this interface, which should never receive a superior BPDU. SW1 is not going to believe these superior BPDUs that SW3 is now the root bridge.)
(##We can see that on SW2 we have one inconsistent port, and it is Gig 0/1. Gig 0/1 is in a Root Inconsistent state because we are receiving superior BPDUs off of this interface, which should never receive a superior BPDU. SW2 is not going to believe these superior BPDUs that SW3 is now the root bridge.)

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